

OptiGrid

Energy Intelligence Platform

User Manual

Version 1.0

July 30, 2026

Team Coreflow



COS 301 Capstone Project

University of Pretoria

Contents

Document Revision History	3
1 Introduction	4
1.1 What is OptiGrid?	4
1.2 Who Should Read This Manual?	4
1.3 Key Features at a Glance	4
1.4 How to Use This Manual	4
2 Getting Started	4
2.1 System Requirements	4
2.2 Accessing the Platform	5
2.3 Creating an Account	5
2.4 Logging In	6
2.5 Logging Out	7
3 Understanding User Roles	7
4 Navigating the Interface	7
4.1 The Main Dashboard	8
4.2 The Navigation Menu	8
4.3 Switching Between Light and Dark Mode	8
5 Working with Buildings	9
5.1 Viewing Your Buildings	9
5.2 Adding a New Building	9
5.3 Viewing Building Details	10
5.4 Editing a Building	10
5.5 Deleting a Building	11
6 Managing Sensors	11
6.1 Viewing a Building's Sensors	11
6.2 Registering a New Sensor	11
6.3 Viewing Sensor Details	12
7 Monitoring Energy Usage	12
7.1 Live Energy Monitoring	12
7.2 Reading the Dashboard Charts	12
7.3 Understanding the Key Figures	13
8 Comparing Buildings	13
8.1 Selecting Buildings to Compare	13

9	Energy Demand Forecasting	13
9.1	Running a Forecast	13
9.2	Reading the Forecast Chart	14
9.3	Understanding Forecast Accuracy	14
10	Manager Features	14
10.1	Viewing Your Assigned Buildings	15
10.2	Filtering and Sorting Buildings	15
11	Administrator Features	15
11.1	Managing All Buildings	15
11.2	Managing Users	16
12	Account Settings	16
12.1	Updating Your Profile	16
13	Troubleshooting	17
14	Frequently Asked Questions	17
15	Getting Help and Support	17
15.1	Contacting Support	17

Document Revision History

Version	Date	Description of Change	Author
1.0	July 30, 2026	Initial release of the OptiGrid User Manual.	Team Coreflow

1 Introduction

1.1 What is OptiGrid?

OptiGrid is a centralized, intelligent energy management system designed to help organizations monitor, analyze, and optimize their energy consumption. By integrating data ingestion, predictive analytics, and optimization insights, OptiGrid empowers users to improve operational efficiency, reduce energy costs, and enhance sustainability across their building portfolios.

1.2 Who Should Read This Manual?

This manual is intended for all users of the OptiGrid platform, including:

- **General Users** who monitor energy usage, view dashboards, analyze reports, and manage their personal account settings.
- **New Users** who require guidance to navigate the OptiGrid platform.
- **Building Managers** who are responsible for the operational management of specific buildings.
- **Administrators** who oversee user access, building profiles, and system configuration.

1.3 Key Features at a Glance

- **Energy Monitoring Dashboard:** Real-time and historical visualization of energy consumption.
- **Energy Demand Forecasting:** Predictive analytics for future energy needs.
- **Building Comparison:** analysis of energy performance across multiple buildings.
- **Sensor Management:** registering and monitoring IoT sensors within buildings.
- **Role-Based Access:** Secure access controls tailored to different user roles.

1.4 How to Use This Manual

This manual is structured to guide you through the OptiGrid platform from initial setup to advanced features. We recommend new users start with the "Getting Started" section to set up their account and learn the basics.

2 Getting Started

2.1 System Requirements

OptiGrid is a web-based platform that runs in web browsers.

- **Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.
- **Internet Connection:** A stable internet connection for optimal performance.
- **Devices:** The platform is accessible on desktop computers and laptops.

2.2 Accessing the Platform

To access the OptiGrid platform, open your web browser and navigate to the official website: <https://www.optigrid.co.za/>.

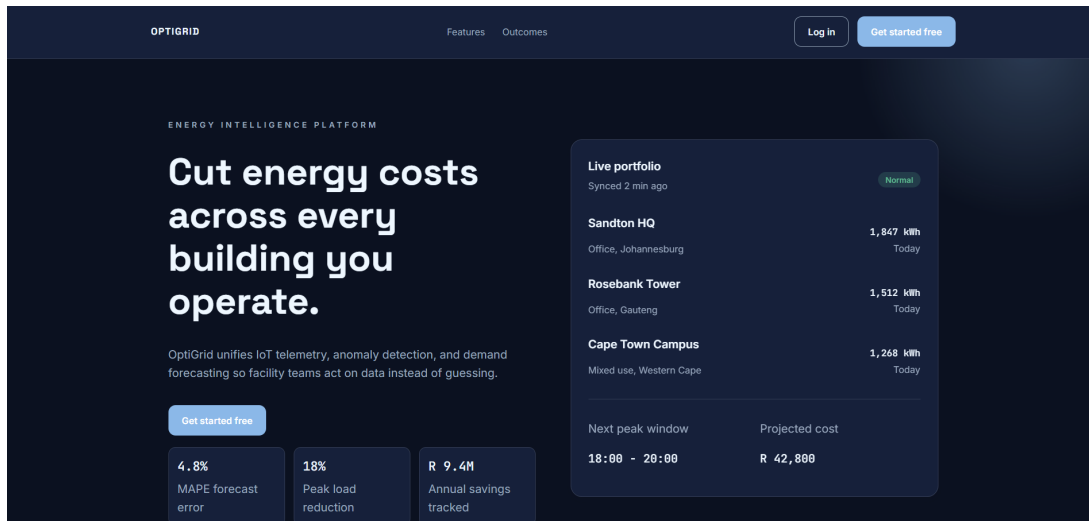


Figure 1: The OptiGrid Landing Page

The landing page introduces the OptiGrid platform and provides options to log in to an existing account or create a new one.

2.3 Creating an Account

If you are a new user, you must create an account to access the platform. Follow these steps to register:

1. On the landing page, locate and click the "Sign Up" or "Register" button.
2. Fill in the registration form with the required information:
 - First Name
 - Last Name
 - Email Address
 - Password (ensure it meets the security requirements)
3. Click the "Create Account" button to complete the registration.

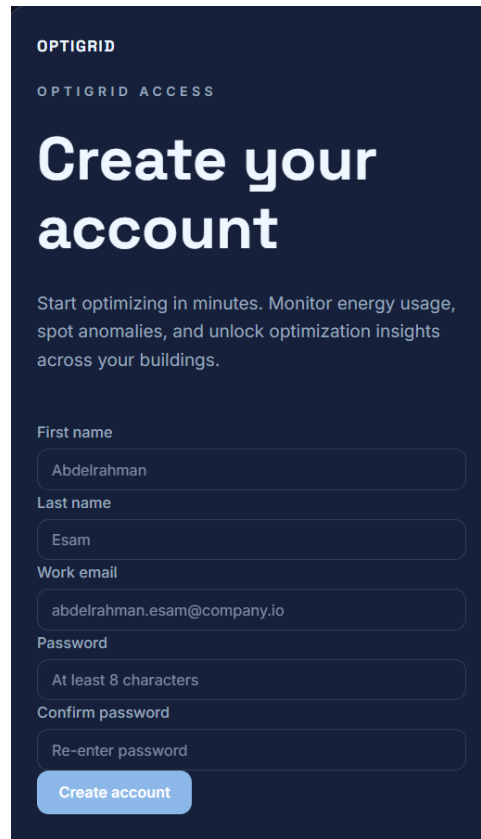
The image shows a dark-themed web form for creating an account. At the top, it says 'OPTIGRID' and 'OPTIGRID ACCESS'. The main heading is 'Create your account'. Below this is a subtext: 'Start optimizing in minutes. Monitor energy usage, spot anomalies, and unlock optimization insights across your buildings.' The form contains several input fields: 'First name' with the value 'Abdelrahman', 'Last name' with 'Esam', 'Work email' with 'abdelrahman.esam@company.io', 'Password' with a placeholder 'At least 8 characters', and 'Confirm password' with a placeholder 'Re-enter password'. A blue 'Create account' button is at the bottom.

Figure 2: The Account Registration Page

2.4 Logging In

Once your account is created, you can log in to the platform:

1. Go to the OptiGrid landing page.
2. Click the "Log In" button.
3. Enter your registered email address and password.
4. Click the "Log In" button to access the dashboard.

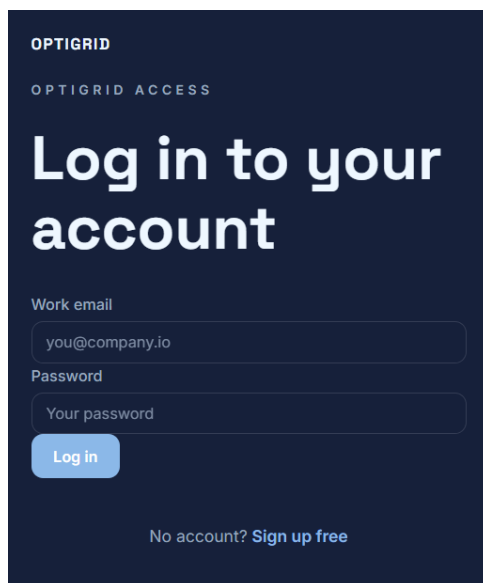


Figure 3: The Login Page

2.5 Logging Out

You can log out using either of the following methods:

- **Navigation Bar:** Click the "Logout" button located at the top right corner of the navigation bar.
- **Settings Page:** Navigate to the settings page and click the "Logout" button at the bottom of the page.

3 Understanding User Roles

OptiGrid uses a role-based access control system to manage permissions and ensure that users have appropriate access to features and data.

Table 1: User Roles and Capabilities

Role	Capabilities
Viewer	View dashboards and energy usage data. Receive basic reports. Cannot make changes to system configurations or data.
Building Manager	Monitor and manage energy consumption for assigned buildings. Register sensors. View forecast data. Generate reports.
Administrator	Full system access. Manage building profiles, user accounts, and role assignments. Configure IoT sensor connections. View and manage all data across the organization.

4 Navigating the Interface

4.1 The Main Dashboard

The main dashboard provides an overview of your energy data. Key elements include:

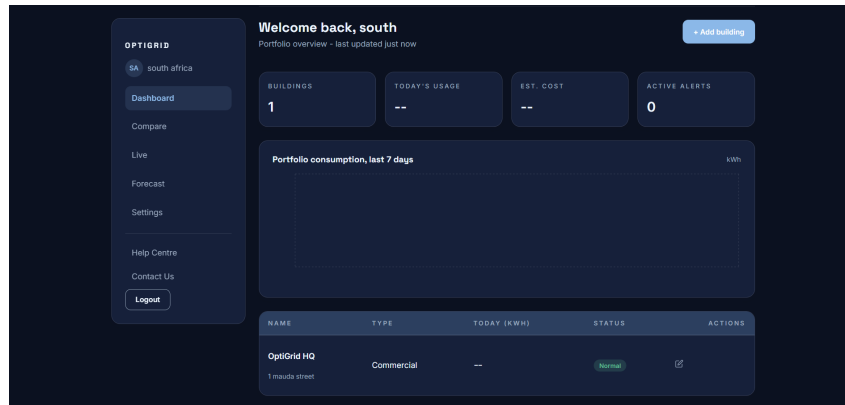


Figure 4: The Main OptiGrid Dashboard

- **Energy Usage per Building:** shows energy usage of each building.
- **Building list:** shows a list of all your buildings
- **Actions:** Buttons to edit and add new buildings or navigate to specific features.

4.2 The Navigation Menu

The navigation menu, typically located at the side of the page, provides access to the core areas of the OptiGrid platform. The specific items available will depend on your user role.

- **Dashboard:** The main overview page for all users.
- **Manage:** Access to building and sensor management features (Building Managers and Administrators).
- **Settings:** Manage account settings and system display.
- **Help:** Access to documentation, FAQs, and support.
- **live readings:** shows the live readings of the buildings.
- **contact us:** ability to contact the team for any issues.
- **Logout:** End your current session.

4.3 Switching Between Light and Dark Mode

To change the visual theme of the platform, navigate to the settings page and locate the theme section. Here you can toggle between light mode and dark mode.

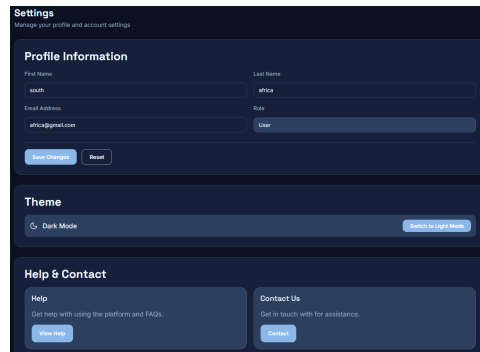


Figure 5: Theme

5 Working with Buildings

5.1 Viewing Your Buildings

To view your buildings, navigate to the dashboard. The building list displays information for each building, such as its name, type, total energy consumption in kilowatt-hours (kWh), and its current operational status.

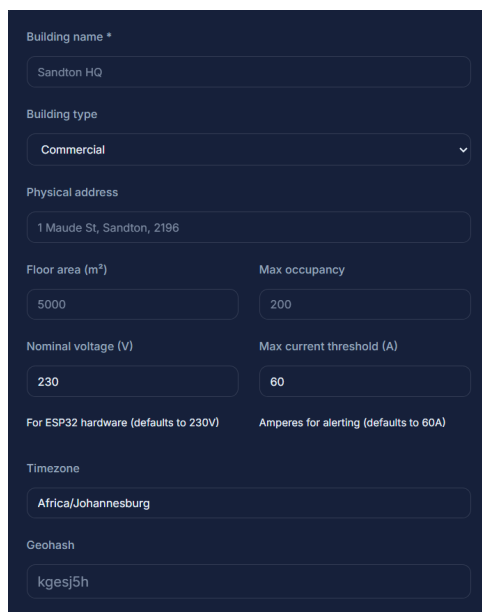
NAME	TYPE	TODAY (KWH)	STATUS	ACTIONS
SWW sef2qe	Commercial	--	Normal	
OptiGrid HQ 1 maude street	Commercial	--	Normal	

Figure 6: List of Buildings

5.2 Adding a New Building

To add a new building to the platform:

1. On the dashboard, click the "Add Building" button.
2. Fill in the building details form:
 - **Building Name:** A name for the building.
 - **Building Type:** The primary function of the building (e.g., Office, Retail, Residential).
 - **Physical Address:** The complete address of the building.
3. Click the "create building" button to add the building.



Building name *

Sandton HQ

Building type

Commercial

Physical address

1 Maude St, Sandton, 2196

Floor area (m²)

5000

Max occupancy

200

Nominal voltage (V)

230

Max current threshold (A)

60

For ESP32 hardware (defaults to 230V)

Amperes for alerting (defaults to 60A)

Timezone

Africa/Johannesburg

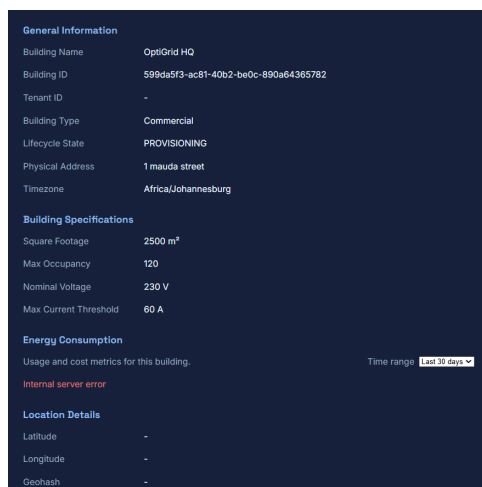
Geohash

kgesj5h

Figure 7: Adding a New Building

5.3 Viewing Building Details

To view detailed information about a specific building, click on its card in the building list. This page provides a view of the building's important details.



General Information

Building Name	OptiGrid HQ
Building ID	599da5f3-ac81-40b2-be0c-890a64365782
Tenant ID	-
Building Type	Commercial
Lifecycle State	PROVISIONING
Physical Address	1 maude street
Timezone	Africa/Johannesburg

Building Specifications

Square Footage	2500 m ²
Max Occupancy	120
Nominal Voltage	230 V
Max Current Threshold	60 A

Energy Consumption

Usage and cost metrics for this building.

Internal server error

Time range: Last 30 days

Location Details

Latitude	-
Longitude	-
Geohash	-

Figure 8: Detailed Building View

5.4 Editing a Building

To edit a building's information, find the edit icon (like a pencil symbol) next to the building's entry in the list. You can then update any of the building details, such as its name, type, or address.

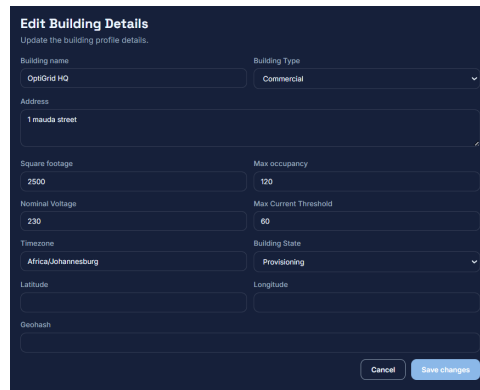


Figure 9: Editing Building Information

5.5 Deleting a Building

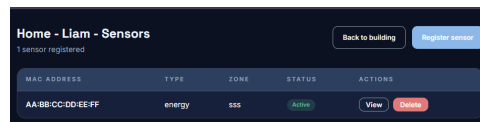
To delete a building, locate the delete icon (typically a trash can symbol) next to the building's entry in the list. You will be prompted to confirm the deletion.

Warning: Deleting a building is a permanent action and cannot be undone.

6 Managing Sensors

6.1 Viewing a Building's Sensors

Building Managers can view a building's sensors by navigating to the manage page and selecting the sensors tab. This list shows all registered sensors, their mac address, their current status, and the data they are collecting.



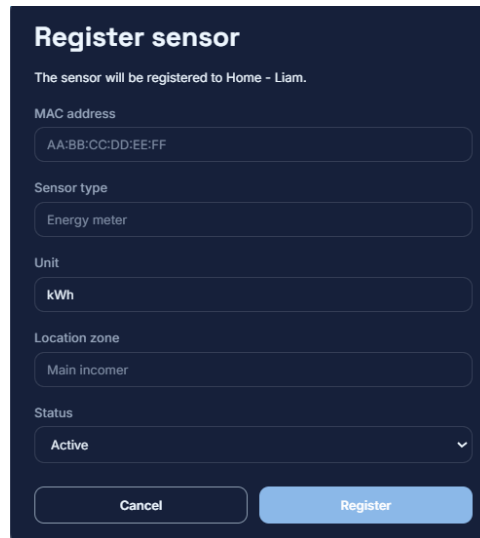
MAC ADDRESS	TYPE	ZONE	STATUS	ACTIONS
A:BB:CC:DD:EE:FF	energy	555	Active	View Delete

Figure 10: Viewing a Building's Sensors

6.2 Registering a New Sensor

To add a new sensor to a building:

1. Navigate to the manage page.
2. Click the "Register Sensor" button.
3. Fill in the sensor registration form:
 - **MAC Address:** The unique hardware identifier of the sensor. This must follow the standard MAC address format.
 - **Sensor Type:** The kind of sensor being registered.
 - **Location Zone:** The specific area within the building where the sensor is installed.
4. Click the "Register" button to complete the process.



Register sensor

The sensor will be registered to Home - Liam.

MAC address
AA:BB:CC:DD:EE:FF

Sensor type
Energy meter

Unit
kWh

Location zone
Main incomer

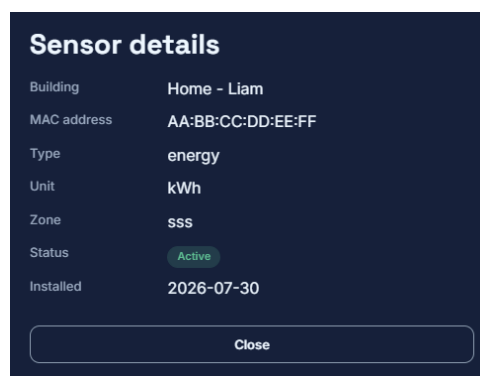
Status
Active

Cancel Register

Figure 11: Registering a New Sensor

6.3 Viewing Sensor Details

To view details for a specific sensor, click the "View" button next to the sensor's entry in the list. This page shows all the important details of a specific sensor.



Sensor details

Building	Home - Liam
MAC address	AA:BB:CC:DD:EE:FF
Type	energy
Unit	kWh
Zone	SSS
Status	Active
Installed	2026-07-30

Close

Figure 12: Viewing Sensor Details

7 Monitoring Energy Usage

The OptiGrid platform provides several powerful tools for monitoring energy consumption.

7.1 Live Energy Monitoring

The live energy monitoring feature provides real-time visualization of energy consumption. This allows users to see current usage and identify immediate changes in demand.

7.2 Reading the Dashboard Charts

The dashboard charts are designed to provide a quick and clear understanding of energy data.

- **Line Charts:** Show trends in energy consumption over time (e.g., hourly, daily, monthly).

7.3 Understanding the Key Figures

The dashboard displays several key figures to help you quickly gauge energy performance:

- **kWh:** Kilowatt-hours, the standard unit of measurement for energy consumption.
- **Energy Cost:** An estimated cost based on the local utility rate and energy consumption.
- **Peak Usage:** The highest energy consumption recorded during a specific period.

8 Comparing Buildings

8.1 Selecting Buildings to Compare

To compare buildings:

1. On the dashboard, navigate to "Compare" page.
2. Choose the two buildings you wish to compare from your building list.
3. Select a time range for the comparison.
4. Choose the metric to analyze.
5. Click the "Compare" button to view the results.

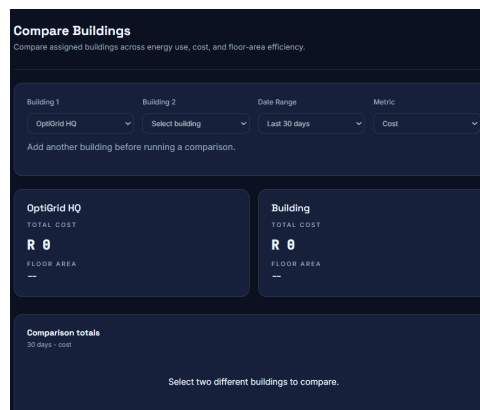


Figure 13: Comparing Building Energy Usage

9 Energy Demand Forecasting

The forecasting feature uses predictive analytics to estimate future energy demand.

9.1 Running a Forecast

To run a forecast:

1. Navigate to the forecast page from the dashboard.
2. Select the building you want to analyze.
3. Click the "Run Forecast" button to generate a prediction.

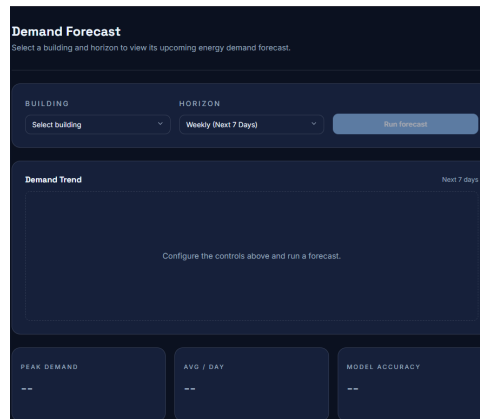


Figure 14: The Energy Demand Forecast Page

9.2 Reading the Forecast Chart

Once a forecast is generated, the results are displayed in a chart. This chart includes:

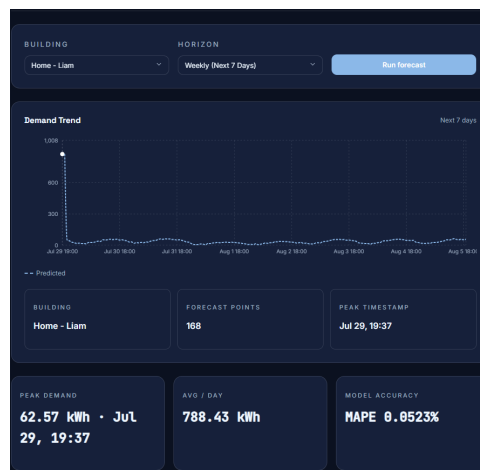


Figure 15: Forecast Results and Chart

- **Demand Trend Graph:** Shows the predicted future demand.
- **Peak Demand:** The highest predicted demand.
- **Average kWh per Day:** The average energy consumption forecasted per day.
- **Model Accuracy:** An indication of the forecast model's reliability.

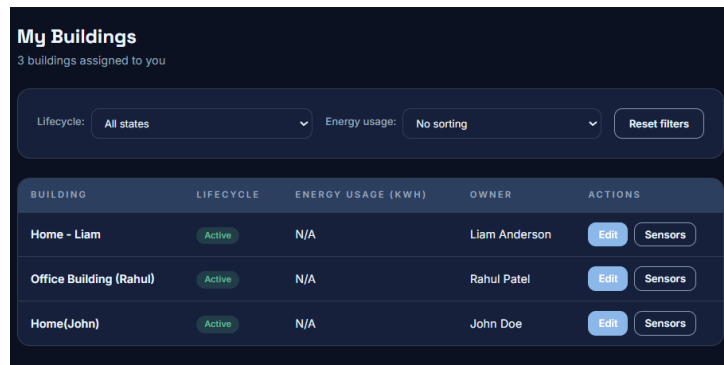
9.3 Understanding Forecast Accuracy

The forecast accuracy metric helps you understand the reliability of the prediction. A higher percentage indicates a more confident forecast.

10 Manager Features

10.1 Viewing Your Assigned Buildings

Upon logging in as a Building Manager, you will see a view specifically focused on the buildings assigned to you.



My Buildings
3 buildings assigned to you

Lifecycle: All states Energy usage: No sorting Reset filters

BUILDING	LIFECYCLE	ENERGY USAGE (KWH)	OWNER	ACTIONS
Home - Liam	Active	N/A	Liam Anderson	Edit Sensors
Office Building (Rahul)	Active	N/A	Rahul Patel	Edit Sensors
Home(John)	Active	N/A	John Doe	Edit Sensors

Figure 16: Manager View of Assigned Buildings

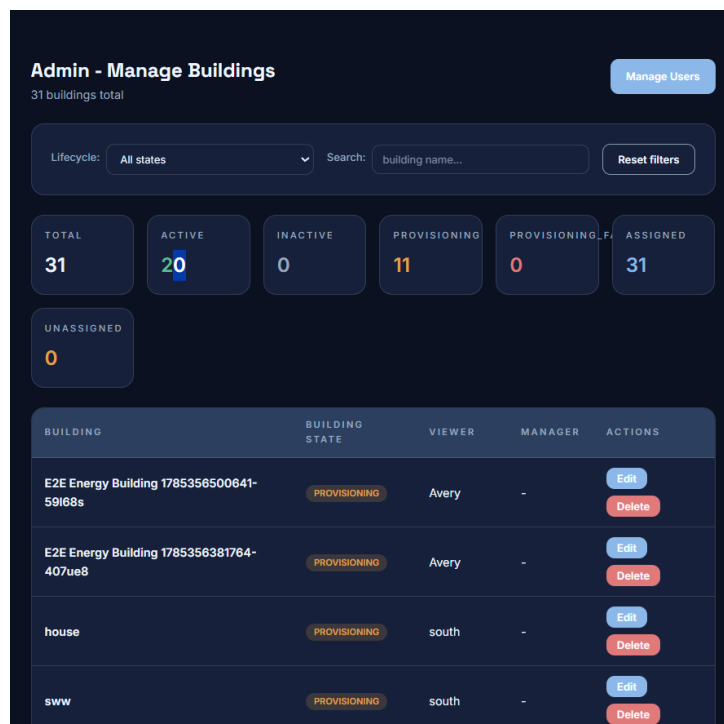
10.2 Filtering and Sorting Buildings

Managers can filter and sort their building list to quickly find the information they need using life cycle and energy usage.

11 Administrator Features

11.1 Managing All Buildings

Administrators can view, edit, and delete all buildings within optigrid. They can also see total buildings in different categories.



Admin - Manage Buildings
31 buildings total

Manage Users

Lifecycle: All states Search: building name... Reset filters

TOTAL	ACTIVE	INACTIVE	PROVISIONING	PROVISIONING_LF	ASSIGNED
31	20	0	11	0	31

UNASSIGNED: 0

BUILDING	BUILDING STATE	VIEWER	MANAGER	ACTIONS
E2E Energy Building 1785356500641-59168s	PROVISIONING	Avery	-	Edit Delete
E2E Energy Building 1785356381764-407ue8	PROVISIONING	Avery	-	Edit Delete
house	PROVISIONING	south	-	Edit Delete
sww	PROVISIONING	south	-	Edit Delete

Figure 17: Administrator View of All Buildings

11.2 Managing Users

The user management feature allows Administrators to view all users, assign user roles, and manage building assignments.

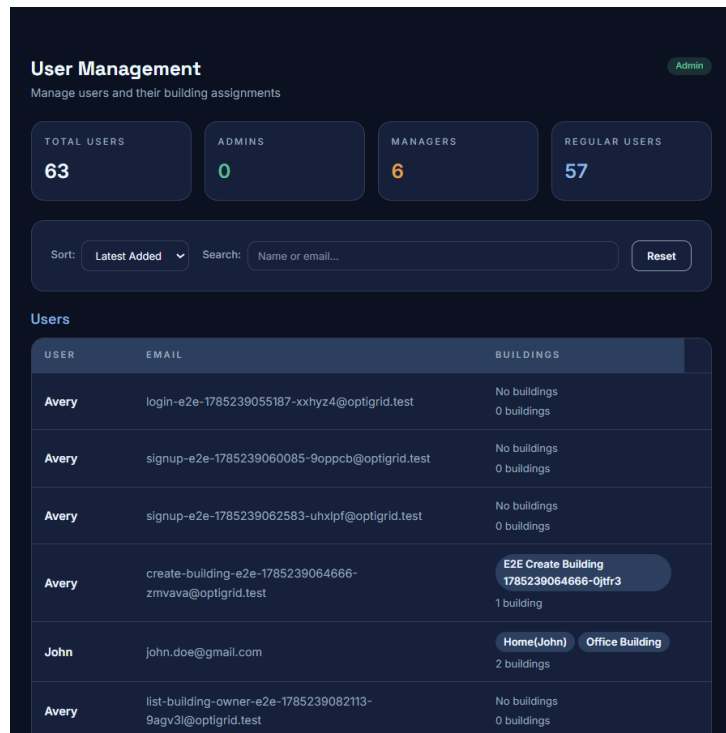


Figure 18: Managing User Accounts and Permissions

12 Account Settings

All users can manage their personal account preferences.

12.1 Updating Your Profile

To update your profile:

1. Navigate to the settings page.
2. Locate the profile section.
3. Update your information, such as your name and email address
4. Click the "Save changes" button to apply the changes.

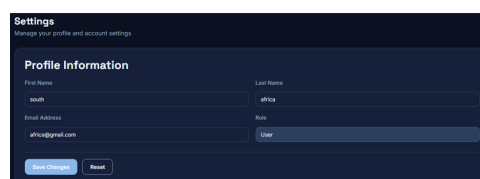


Figure 19: Updating User Profile

13 Troubleshooting

This section addresses common issues and provides solutions.

Table 2: Common Issues and Solutions

Problem	Solution
Cannot log in.	Verify your email and password. Use the "Forgot Password" link to reset your password if needed.
No data is showing for a building.	Check the sensors' status to ensure they are connected and reporting.
The dashboard is loading slowly.	Check your internet connection. Try refreshing the browser.

14 Frequently Asked Questions

The Frequently Asked Questions (FAQ) page provides answers to common questions about using the OptiGrid platform.

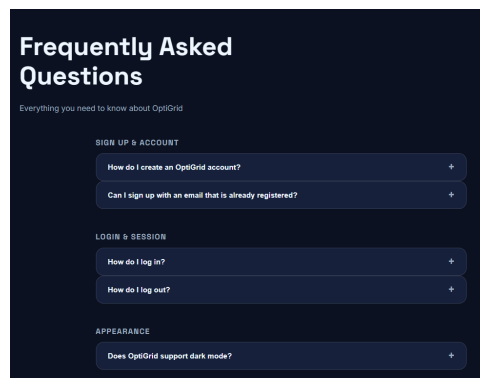
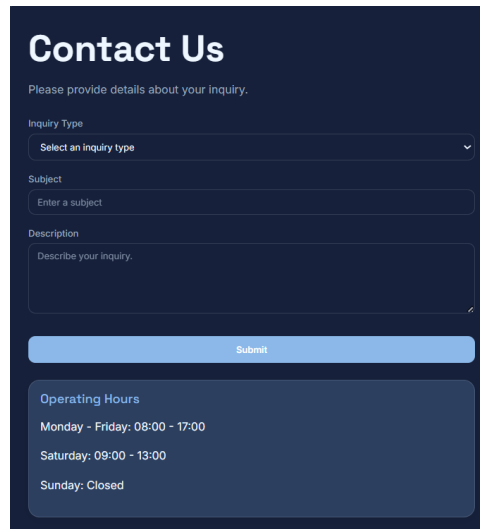


Figure 20: OptiGrid Frequently Asked Questions

15 Getting Help and Support

15.1 Contacting Support

If you encounter an issue not covered in this manual or the FAQ, you can contact the OptiGrid support team directly.

A dark-themed contact form titled "Contact Us". Below the title is a subtitle "Please provide details about your inquiry." The form contains three input fields: "Inquiry Type" (a dropdown menu with the placeholder "Select an inquiry type"), "Subject" (a text box with the placeholder "Enter a subject"), and "Description" (a larger text box with the placeholder "Describe your inquiry."). Below these fields is a blue "Submit" button. At the bottom of the form is a section titled "Operating Hours" which lists: "Monday - Friday: 08:00 - 17:00", "Saturday: 09:00 - 13:00", and "Sunday: Closed".

Contact Us

Please provide details about your inquiry.

Inquiry Type
Select an inquiry type

Subject
Enter a subject

Description
Describe your inquiry.

Submit

Operating Hours
Monday - Friday: 08:00 - 17:00
Saturday: 09:00 - 13:00
Sunday: Closed

Figure 21: Contacting Support

To submit a support request:

1. Navigate to the help or support page.
2. Select the appropriate inquiry type from the dropdown menu.
3. Enter a subject that clearly describes the issue.
4. Provide a detailed description of the problem in the text box.
5. Click the "Submit" button to send your request.

.